

# INTEGRAL SOLUTIONS OF $Z^2 - XY^2 = X^3 + 2$

**ABSTRACT.** We give a complete and effective description of the integer solutions of  $z^2 - xy^2 = x^3 + 2$ . The nonpositive and positive even values of  $x$  are settled completely, while positive square values of  $x$  reduce to a finite divisor factorization. For every positive nonsquare value of  $x$ , the solutions split uniquely into finitely many orbits under the positive norm-one units of  $\mathbb{Z}[\sqrt{x}]$ , and the orbit representatives lie in an explicit bounded interval. In particular, a single value,  $x = 23$ , already gives infinitely many integer solutions.

## 1. STATEMENT OF THE CLASSIFICATION

We study the equation

$$(1.1) \quad z^2 - xy^2 = x^3 + 2, \quad (x, y, z) \in \mathbb{Z}^3.$$

The equation is unchanged by either of the independent sign changes  $y \mapsto -y$  and  $z \mapsto -z$ . It is therefore natural to classify the pairs  $(|y|, |z|)$  and restore signs at the end.

Define

$$(1.2) \quad \mathcal{A} = \{n \in \mathbb{Z}_{>0} : n \text{ is odd and every prime } p \mid n \text{ satisfies } p \equiv 1 \text{ or } 7 \pmod{8}\}.$$

The integer 1 belongs to  $\mathcal{A}$ , since the condition on its prime divisors is vacuous.

For a positive nonsquare integer  $D$ , let

$$(1.3) \quad \varepsilon_D = u_D + v_D\sqrt{D} > 1$$

be the least real number of the indicated form for which

$$(1.4) \quad u_D, v_D \in \mathbb{Z}, \quad u_D^2 - Dv_D^2 = 1.$$

We shall prove from first principles that  $\varepsilon_D$  exists, that  $u_D, v_D > 0$ , and that every positive norm-one unit in  $\mathbb{Z}[\sqrt{D}]$  is a nonnegative power of  $\varepsilon_D$ .

Put

$$(1.5) \quad N_D = D^3 + 2$$

and define the reduced seed set

$$(1.6) \quad \mathcal{R}_D = \left\{ (A, B) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0} : \begin{array}{l} A^2 - DB^2 = N_D, \\ \sqrt{N_D} \leq A + B\sqrt{D} < \varepsilon_D\sqrt{N_D} \end{array} \right\}.$$

**Theorem 1.1** (Complete classification). *All integer solutions of (1.1) are exactly the following.*

(i) *The solutions with  $x \leq 0$  are*

$$(1.7) \quad (-1, 0, 1), \quad (-1, 0, -1), \quad (-1, 1, 0), \quad (-1, -1, 0).$$

*There are no solutions with  $x = 0$  or  $x \leq -2$ .*

(ii) *Let  $m \in \mathcal{A}$ , put  $x = m^2$ , and set*

$$(1.8) \quad N_m = m^6 + 2.$$

*For each positive divisor  $d \mid N_m$  satisfying*

$$(1.9) \quad d < \sqrt{N_m}, \quad \frac{N_m}{d} \equiv d \pmod{2m},$$

*put*

$$(1.10) \quad Y_{m,d} = \frac{N_m/d - d}{2m}, \quad Z_{m,d} = \frac{N_m/d + d}{2}.$$

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2020 *Mathematics Subject Classification.* Primary 11D09; Secondary 11D25, 11D41.

*Key words and phrases.* Diophantine equation, Pell equation, generalized Pell equation, integral points, quadratic norm.

Then

$$(1.11) \quad (x, y, z) = (m^2, \sigma Y_{m,d}, \tau Z_{m,d}), \quad \sigma, \tau \in \{1, -1\},$$

are precisely the solutions for which  $x$  is a positive square. For fixed  $m$ , the divisor  $d$  is uniquely determined by  $(|y|, |z|)$ .

(iii) Let  $D \in \mathcal{A}$  be nonsquare. For a seed  $(A_0, B_0) \in \mathcal{R}_D$ , define integer sequences by

$$(1.12) \quad \begin{aligned} A_{n+1} &= u_D A_n + D v_D B_n, \\ B_{n+1} &= v_D A_n + u_D B_n \end{aligned} \quad (n \geq 0).$$

Equivalently,

$$(1.13) \quad A_n + B_n \sqrt{D} = (A_0 + B_0 \sqrt{D}) \varepsilon_D^n.$$

Then

$$(1.14) \quad (x, y, z) = (D, \sigma B_n, \tau A_n), \quad (A_0, B_0) \in \mathcal{R}_D, \quad n \in \mathbb{Z}_{\geq 0}, \quad \sigma, \tau \in \{1, -1\},$$

are precisely the solutions for which  $x$  is positive and nonsquare. Every pair  $(|z|, |y|)$  has a unique representation (1.13). Moreover,  $\mathcal{R}_D$  is finite and every one of its elements satisfies

$$(1.15) \quad 0 \leq B_0 < v_D \sqrt{N_D}, \quad 0 < A_0 < u_D \sqrt{N_D}.$$

There are no further solutions. In particular, if  $x > 0$  occurs in a solution, then  $x \in \mathcal{A}$ .

The theorem is an effective classification. In the square case, one tests the finitely many divisors of  $m^6 + 2$ . In the nonsquare case, one first finds the least Pell unit  $\varepsilon_D$ , and then the strict bound  $B_0^2 < v_D^2 N_D$  leaves only finitely many possible integers  $B_0$  to check in (1.6).

## 2. RESTRICTIONS ON THE PARAMETER $x$

We begin with the nonpositive values.

**Lemma 2.1.** *The only solutions of (1.1) with  $x \leq 0$  are those in (1.7).*

*Proof.* If  $x = 0$ , then (1.1) becomes  $z^2 = 2$ , which has no integer solution.

Now let  $x = -a$ , where  $a \in \mathbb{Z}_{>0}$ . Substitution in (1.1) gives

$$(2.1) \quad z^2 + ay^2 = 2 - a^3.$$

If  $a \geq 2$ , then the left-hand side is nonnegative whereas  $2 - a^3 \leq -6$ , a contradiction. Hence  $a = 1$ , so that  $x = -1$  and

$$(2.2) \quad y^2 + z^2 = 1.$$

Since  $y^2, z^2$  are nonnegative integers whose sum is 1, one of them is 1 and the other is 0. This gives exactly the four triples in (1.7).  $\square$

The next elementary residue lemma is the only quadratic-residue input needed below.

**Lemma 2.2.** *Let  $p$  be an odd prime. If the congruence*

$$(2.3) \quad r^2 \equiv 2 \pmod{p}$$

*has an integer solution, then*

$$(2.4) \quad p \equiv 1 \quad \text{or} \quad 7 \pmod{8}.$$

*Proof.* Write  $h = (p-1)/2$ . For each  $j \in \{1, \dots, h\}$ , reduce  $2j$  modulo  $p$  to a signed representative  $\delta_j a_j$ , where  $\delta_j \in \{1, -1\}$  and  $a_j \in \{1, \dots, h\}$ . The numbers  $a_1, \dots, a_h$  form a permutation of  $1, \dots, h$ . Indeed, if  $a_i = a_j$ , then  $2i \equiv \pm 2j \pmod{p}$ . The plus sign gives  $i = j$ , whereas the minus sign would give  $i + j \equiv 0 \pmod{p}$ , which is impossible because  $2 \leq i + j \leq p-1$ .

Let  $M$  be the number of negative signs  $\delta_j = -1$ . Multiplying all of the congruences  $2j \equiv \delta_j a_j \pmod{p}$ , and then cancelling the nonzero residue  $h!$ , yields

$$(2.5) \quad 2^h \equiv (-1)^M \pmod{p}.$$

Because  $2j$  lies between 2 and  $p-1$ , the signed representative is negative precisely when  $2j > p/2$ . Therefore

$$(2.6) \quad M = h - \left\lfloor \frac{p}{4} \right\rfloor.$$

A direct calculation in the four odd residue classes modulo 8 gives

$$(2.7) \quad M \equiv \begin{cases} 0 & (\text{mod } 2), \quad p \equiv 1, 7 \pmod{8}, \\ 1 & (\text{mod } 2), \quad p \equiv 3, 5 \pmod{8}. \end{cases}$$

Suppose now that (2.3) holds. Then  $p \nmid r$ . Multiplication by  $r$  permutes the nonzero residue classes modulo  $p$ , so

$$r^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Cancelling  $(p-1)!$  proves  $r^{p-1} \equiv 1 \pmod{p}$ . Consequently

$$(2.8) \quad 2^h \equiv (r^2)^h = r^{p-1} \equiv 1 \pmod{p}.$$

Equations (2.5)–(2.7) now force  $M$  to be even, and hence  $p \equiv 1$  or  $7 \pmod{8}$ .  $\square$

**Proposition 2.3.** *If  $(x, y, z)$  satisfies (1.1) with  $x > 0$ , then  $x$  is odd and  $x \in \mathcal{A}$ . In particular, no positive even value of  $x$  occurs.*

*Proof.* Reduction of (1.1) modulo  $x$  gives

$$(2.9) \quad z^2 \equiv 2 \pmod{x}.$$

Thus, for every odd prime  $p \mid x$ , the congruence  $z^2 \equiv 2 \pmod{p}$  holds. Lemma 2.2 then gives  $p \equiv 1$  or  $7 \pmod{8}$ .

It remains to exclude even  $x$ . If  $4 \mid x$ , then (2.9) implies  $z^2 \equiv 2 \pmod{4}$ , impossible because a square modulo 4 is 0 or 1. Hence any hypothetical even solution would have

$$(2.10) \quad x = 2m$$

with  $m$  odd. Equation (1.1) becomes

$$(2.11) \quad z^2 = 2my^2 + 8m^3 + 2.$$

If  $y$  were even, then the right-hand side would be congruent to 2 modulo 8, again impossible for a square. Thus  $y$  is odd. Every odd square is congruent to 1 or 9 (mod 16), and therefore  $2my^2 \equiv 2m \pmod{16}$ . Since  $m$  is odd, also  $8m^3 \equiv 8 \pmod{16}$ . It follows that

$$(2.12) \quad z^2 \equiv 2m + 10 \pmod{16}.$$

The right-hand side is even, so it must be one of the even square residues  $0, 4 \pmod{16}$ . Hence

$$(2.13) \quad m \equiv 3 \quad \text{or} \quad 5 \pmod{8}.$$

On the other hand, reducing (2.11) modulo  $m$  gives  $z^2 \equiv 2 \pmod{m}$ . By the first paragraph of the proof, every prime divisor of  $m$  is congruent to 1 or  $-1 \pmod{8}$ . The product of arbitrarily many residues from  $\{1, -1\}$  is again 1 or  $-1$ , even when prime powers are counted with multiplicity. Therefore

$$(2.14) \quad m \equiv 1 \quad \text{or} \quad 7 \pmod{8},$$

contradicting (2.13). Thus  $x$  is odd. Together with the prime restriction already proved, this says exactly that  $x \in \mathcal{A}$ .  $\square$

## 3. THE POSITIVE SQUARE CASE

We now give a finite factorization of all solutions for which  $x$  is a square.

**Proposition 3.1.** *Let  $m \in \mathbb{Z}_{>0}$ . Equation (1.1) has a solution with  $x = m^2$  if and only if  $m \in \mathcal{A}$  and there is a positive divisor  $d \mid m^6 + 2$  satisfying (1.9). For every such  $d$ , all corresponding solutions are given by (1.10)–(1.11), and this parametrization is unique up to the independent signs of  $y$  and  $z$ .*

*Proof.* Suppose first that  $(m^2, y, z)$  is a solution. By Proposition 2.3,  $m^2 \in \mathcal{A}$ , and hence  $m \in \mathcal{A}$ . In particular,  $m$  is odd. The equation is

$$(3.1) \quad z^2 - m^2 y^2 = m^6 + 2 = N_m.$$

The integer  $N_m$  is not a square, since

$$(3.2) \quad m^6 < N_m = m^6 + 2 < (m^3 + 1)^2 = m^6 + 2m^3 + 1.$$

In particular,  $y \neq 0$ , for otherwise (3.1) would make  $N_m = z^2$ .

Set

$$(3.3) \quad Y = |y|, \quad Z = |z|.$$

Since  $Z^2 - m^2 Y^2 = N_m > 0$ , we have  $Z > mY$ . Therefore

$$(3.4) \quad d = Z - mY, \quad e = Z + mY$$

are positive integers satisfying

$$(3.5) \quad de = N_m, \quad d < e, \quad e - d = 2mY.$$

Thus  $d < \sqrt{N_m}$ ,  $e = N_m/d$ , and

$$(3.6) \quad \frac{N_m}{d} = e \equiv d \pmod{2m}.$$

Solving (3.4) for  $Y, Z$  gives

$$(3.7) \quad Y = \frac{e - d}{2m} = \frac{N_m/d - d}{2m}, \quad Z = \frac{e + d}{2} = \frac{N_m/d + d}{2}.$$

This proves the necessity of the stated formula. It also proves uniqueness: from  $(Y, Z)$ , the divisor is forced to be  $d = Z - mY$ .

Conversely, suppose  $m \in \mathcal{A}$  and  $d \mid N_m$  satisfies (1.9). Put  $e = N_m/d$ . Since  $m$  is odd,  $N_m, d$ , and  $e$  are all odd. The congruence  $e \equiv d \pmod{2m}$  shows that

$$Y = \frac{e - d}{2m}$$

is an integer, and the oddness of  $d, e$  shows that

$$Z = \frac{e + d}{2}$$

is an integer. The strict inequality  $d < \sqrt{N_m}$  implies  $e > d$ , so  $Y > 0$ . Finally,

$$(3.8) \quad \begin{aligned} Z^2 - m^2 Y^2 &= \frac{(e + d)^2 - (e - d)^2}{4} \\ &= ed = N_m = m^6 + 2. \end{aligned}$$

This is precisely (1.1) with  $x = m^2$ . Independent choices of the signs of  $y$  and  $z$  give all solutions belonging to  $(Y, Z)$ .  $\square$

*Example 3.2.* For  $m = 1$ , one has  $N_m = 3$ , and  $d = 1$  gives

$$(x, y, z) = (1, \pm 1, \pm 2).$$

For  $m = 7$ , one has  $N_m = 7^6 + 2 = 117651$ . Taking  $d = 3$ , so that  $N_m/d = 39217$ , gives

$$Y = \frac{39217 - 3}{14} = 2801, \quad Z = \frac{39217 + 3}{2} = 19610.$$

Hence

$$(x, y, z) = (49, \pm 2801, \pm 19610)$$

is a fourfold sign family of solutions.

#### 4. PELL UNITS AND THE NONSQUARE CASE

For a positive nonsquare  $D$ , write

$$(4.1) \quad N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2.$$

We first prove all of the unit theory that will be used.

**Proposition 4.1** (Positive Pell units). *Let  $D \in \mathbb{Z}_{>0}$  be nonsquare. Then the following statements hold.*

(a) *There exist positive integers  $u, v$  such that*

$$(4.2) \quad u^2 - Dv^2 = 1.$$

(b) *Among all real numbers  $u + v\sqrt{D} > 1$  with  $u, v \in \mathbb{Z}$  and  $u^2 - Dv^2 = 1$ , there is a least one, denoted  $\varepsilon_D = u_D + v_D\sqrt{D}$ . Its coefficients satisfy  $u_D, v_D \in \mathbb{Z}_{>0}$ .*

(c) *If  $\eta = a + b\sqrt{D} > 1$ , where  $a, b \in \mathbb{Z}$  and  $a^2 - Db^2 = 1$ , then there is a unique integer  $n \geq 1$  such that*

$$(4.3) \quad \eta = \varepsilon_D^n.$$

*Together with  $1 = \varepsilon_D^0$ , the nonnegative powers of  $\varepsilon_D$  are exactly the positive norm-one units whose real value is at least 1.*

*Proof.* We first prove existence. For every integer  $T \geq 2$ , consider the  $T + 1$  fractional parts

$$0, \{\sqrt{D}\}, \{2\sqrt{D}\}, \dots, \{T\sqrt{D}\}.$$

Divide  $[0, 1)$  into  $T$  half-open intervals of length  $1/T$ . Two of the displayed fractional parts lie in the same interval. Subtracting the corresponding multiples of  $\sqrt{D}$ , we obtain integers  $r, s$  such that

$$(4.4) \quad 1 \leq s \leq T, \quad |s\sqrt{D} - r| < \frac{1}{T}.$$

Since  $s\sqrt{D} \geq \sqrt{2}$  and  $1/T \leq 1/2$ , every  $r$  satisfying (4.4) is positive. The integer

$$(4.5) \quad k = r^2 - Ds^2$$

is nonzero because  $\sqrt{D}$  is irrational. Moreover,

$$(4.6) \quad \begin{aligned} |k| &= |r - s\sqrt{D}|(r + s\sqrt{D}) \\ &< \frac{1}{T} \left( 2s\sqrt{D} + \frac{1}{T} \right) \\ &\leq 2\sqrt{D} + \frac{1}{T^2} < 2\sqrt{D} + 1. \end{aligned}$$

As  $T$  tends to infinity, the denominators  $s$  obtained in (4.4) cannot remain bounded. Indeed, if  $1 \leq s \leq S$ , then each distance

$$\text{dist}(s\sqrt{D}, \mathbb{Z})$$

is positive; the minimum of these finitely many positive numbers would be a positive constant  $\delta$ , contradicting (4.4) for  $T > 1/\delta$ . Hence we obtain infinitely many distinct pairs  $(r, s)$ .

By (4.6), the nonzero integer  $k = r^2 - Ds^2$  lies in a fixed finite set. Thus one fixed nonzero value of  $k$  occurs for infinitely many pairs. Within this infinite fixed- $k$  subsequence,  $s$  is again unbounded, because for each fixed  $s$  the equation  $r^2 - Ds^2 = k$  permits at most two integers  $r$ . Hence  $r + s\sqrt{D}$  is unbounded. Among those pairs, two, say  $(r_1, s_1)$  and  $(r_2, s_2)$ , have simultaneously

$$(4.7) \quad r_1 \equiv r_2 \pmod{|k|}, \quad s_1 \equiv s_2 \pmod{|k|}.$$

We may label the two pairs so that

$$(4.8) \quad r_1 + s_1\sqrt{D} > r_2 + s_2\sqrt{D} > 0.$$

Consider their quotient. Since both numerator and denominator have norm  $k$ ,

$$(4.9) \quad \frac{r_1 + s_1\sqrt{D}}{r_2 + s_2\sqrt{D}} = \frac{(r_1 + s_1\sqrt{D})(r_2 - s_2\sqrt{D})}{k} = u + v\sqrt{D},$$

where

$$(4.10) \quad u = \frac{r_1r_2 - Ds_1s_2}{k}, \quad v = \frac{s_1r_2 - r_1s_2}{k}.$$

The congruences (4.7) imply that both numerators in (4.10) are divisible by  $|k|$ , and hence  $u, v \in \mathbb{Z}$ . Taking norms in (4.9) gives

$$u^2 - Dv^2 = 1.$$

The quotient in (4.9) is greater than 1 by (4.8). Its conjugate is its positive reciprocal. Therefore

$$u = \frac{\eta + \eta^{-1}}{2} > 0, \quad v = \frac{\eta - \eta^{-1}}{2\sqrt{D}} > 0,$$

where  $\eta$  denotes the quotient. This proves part (a).

Let  $\mathcal{U}_D$  be the nonempty set of all integer-coefficient norm-one units with real value greater than 1. Choose any  $\eta_0 \in \mathcal{U}_D$ . There are only finitely many elements  $u + v\sqrt{D} \in \mathcal{U}_D$  satisfying  $1 < u + v\sqrt{D} \leq \eta_0$ , because then  $0 < v < \eta_0/\sqrt{D}$  and  $0 < u < \eta_0$ . That finite nonempty set has a least element; it is also the least element of all of  $\mathcal{U}_D$ . Denote it by  $\varepsilon_D$ . The same formulas involving a unit and its positive reciprocal show that its coefficients  $u_D, v_D$  are positive. This proves part (b).

Finally, let  $\eta \in \mathcal{U}_D$ . Since  $\varepsilon_D > 1$ , there is a unique integer  $n \geq 0$  such that

$$(4.11) \quad \varepsilon_D^n \leq \eta < \varepsilon_D^{n+1}.$$

The inverse  $\varepsilon_D^{-1} = u_D - v_D\sqrt{D}$  again has integer coefficients and norm 1. Hence

$$\theta = \eta \varepsilon_D^{-n}$$

is an integer-coefficient norm-one unit satisfying

$$(4.12) \quad 1 \leq \theta < \varepsilon_D.$$

If  $\theta > 1$ , then  $\theta \in \mathcal{U}_D$ , contradicting the minimality of  $\varepsilon_D$ . Thus  $\theta = 1$ , and  $\eta = \varepsilon_D^n$ . Since  $\eta > 1$ , one has  $n \geq 1$ . Uniqueness of  $n$  follows from the strict monotonicity of the powers of  $\varepsilon_D > 1$ . This proves part (c).  $\square$

We next prove a general orbit decomposition for a positive generalized Pell equation. This is the exact mechanism behind part (iii) of Theorem 1.1.

**Proposition 4.2** (Finite reduced-orbit decomposition). *Let  $D \in \mathbb{Z}_{>0}$  be nonsquare, let  $N \in \mathbb{Z}_{>0}$ , and let  $\varepsilon_D = u_D + v_D\sqrt{D}$  be as in Proposition 4.1. Define*

$$(4.13) \quad \mathcal{R}(D, N) = \left\{ (A, B) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0} : \begin{array}{l} A^2 - DB^2 = N, \\ \sqrt{N} \leq A + B\sqrt{D} < \varepsilon_D\sqrt{N} \end{array} \right\}.$$

*Then multiplication induces a bijection*

$$(4.14) \quad \mathcal{R}(D, N) \times \mathbb{Z}_{\geq 0} \longrightarrow \left\{ (A, B) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0} : A^2 - DB^2 = N \right\},$$

*where  $((A_0, B_0), n)$  is sent to the coefficients of*

$$(4.15) \quad (A_0 + B_0\sqrt{D})\varepsilon_D^n.$$

*The seed set  $\mathcal{R}(D, N)$  is finite, and every seed satisfies*

$$(4.16) \quad 0 \leq B < v_D\sqrt{N}, \quad 0 < A < u_D\sqrt{N}.$$

*Proof.* Let  $(A, B) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$  satisfy

$$(4.17) \quad A^2 - DB^2 = N.$$

Set

$$(4.18) \quad t = A + B\sqrt{D}.$$

Since  $A > 0$ ,  $B \geq 0$ , and (4.17) has positive right-hand side, one has  $A > B\sqrt{D}$ . Thus both  $t$  and its conjugate  $t' = A - B\sqrt{D}$  are positive, and

$$(4.19) \quad tt' = N.$$

Moreover,  $t \geq t'$ , so  $t^2 \geq tt' = N$ , whence

$$(4.20) \quad t \geq \sqrt{N}.$$

There is a unique integer  $n \geq 0$  for which

$$(4.21) \quad \varepsilon_D^n \leq \frac{t}{\sqrt{N}} < \varepsilon_D^{n+1}.$$

Put

$$(4.22) \quad t_0 = t\varepsilon_D^{-n}.$$

Since  $\varepsilon_D^{-1} = u_D - v_D\sqrt{D}$ , the number  $t_0$  has integer coefficients:

$$(4.23) \quad t_0 = A_0 + B_0\sqrt{D}, \quad A_0, B_0 \in \mathbb{Z}.$$

Its norm is  $N$ , and (4.21) gives

$$(4.24) \quad \sqrt{N} \leq t_0 < \varepsilon_D \sqrt{N}.$$

Its conjugate is  $t'_0 = N/t_0 > 0$ . In addition,  $t_0 \geq \sqrt{N} \geq t'_0$ , so

$$(4.25) \quad A_0 = \frac{t_0 + t'_0}{2} > 0, \quad B_0 = \frac{t_0 - t'_0}{2\sqrt{D}} \geq 0.$$

Thus  $(A_0, B_0) \in \mathcal{R}(D, N)$ , and

$$t = (A_0 + B_0\sqrt{D})\varepsilon_D^n.$$

This proves surjectivity of (4.14).

The integer  $n$  is uniquely determined by (4.21), after which  $t_0 = t\varepsilon_D^{-n}$  is uniquely determined. Hence the representation is unique, proving injectivity.

Conversely, start with a seed  $(A_0, B_0) \in \mathcal{R}(D, N)$ . Because  $\varepsilon_D^n \in \mathbb{Z}[\sqrt{D}]$  and has norm 1, the product (4.15) has integer coefficients  $A_n, B_n$  and norm  $N$ . Since  $A_0 > 0$ ,  $B_0 \geq 0$ , and  $u_D, v_D > 0$ , its coefficients remain in  $\mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$ . Explicitly they obey

$$(4.26) \quad A_{n+1} = u_D A_n + D v_D B_n, \quad B_{n+1} = v_D A_n + u_D B_n.$$

This verifies directly that the map in (4.14) is well-defined.

It remains to establish the bounds and finiteness. For a seed, let  $t = A + B\sqrt{D}$ . Since its conjugate equals  $N/t$ ,

$$(4.27) \quad A = \frac{t + N/t}{2}, \quad B = \frac{t - N/t}{2\sqrt{D}}.$$

On  $[\sqrt{N}, \infty)$ , the first right-hand side is nondecreasing and the second is strictly increasing. Since  $t < \varepsilon_D \sqrt{N}$ , we obtain

$$(4.28) \quad A < \frac{\varepsilon_D + \varepsilon_D^{-1}}{2} \sqrt{N} = u_D \sqrt{N},$$

and

$$(4.29) \quad B < \frac{\varepsilon_D - \varepsilon_D^{-1}}{2\sqrt{D}} \sqrt{N} = v_D \sqrt{N}.$$

Here we used

$$\varepsilon_D^{-1} = u_D - v_D \sqrt{D}.$$

The bounds leave only finitely many integer pairs  $(A, B)$ , so  $\mathcal{R}(D, N)$  is finite.  $\square$

**Proposition 4.3.** *Let  $D > 0$  be nonsquare. The solutions of (1.1) with  $x = D$  are exactly those in (1.14). If there is at least one such solution, then there are infinitely many.*

*Proof.* For  $x = D$ , equation (1.1) is

$$(4.30) \quad z^2 - Dy^2 = D^3 + 2 = N_D.$$

Given a solution, put  $A = |z|$  and  $B = |y|$ . Then  $(A, B) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$  and

$$A^2 - DB^2 = N_D.$$

Proposition 4.2, applied with  $N = N_D$ , gives a unique seed  $(A_0, B_0) \in \mathcal{R}_D$  and a unique  $n \geq 0$  such that

$$A + B\sqrt{D} = (A_0 + B_0\sqrt{D})\varepsilon_D^n.$$

This is exactly the representation in (1.14) after the independent signs of  $y, z$  are restored.

Conversely, each product in (1.13) has norm  $N_D$ , so its coefficients satisfy (4.30); hence every listed triple is a solution. The bounds (1.15) are the specialization of (4.16).

If a seed exists, then the positive real numbers

$$(A_0 + B_0\sqrt{D})\varepsilon_D^n \quad (n \geq 0)$$

are strictly increasing because  $\varepsilon_D > 1$ . Their coefficient pairs are therefore distinct, and so the seed produces infinitely many solutions.  $\square$

*Proof of Theorem 1.1.* Lemma 2.1 proves part (i). Proposition 2.3 shows that every positive solution value of  $x$  belongs to  $\mathcal{A}$  and that no positive even value is possible. Every positive integer is either a square or a nonsquare. Proposition 3.1 gives the complete and unique description in the square case, while Proposition 4.3 and Proposition 4.2 give the complete, unique, and finite seed-orbit description in the nonsquare case. These mutually exclusive cases exhaust all integers  $x$ , so no further solutions exist.  $\square$

## 5. CONSEQUENCES AND EXAMPLES

**Corollary 5.1.** *For each fixed integer  $x$ , the following alternatives hold.*

- (i) *If  $x < -1$ ,  $x = 0$ , or  $x > 0$  is even, there are no solutions.*
- (ii) *If  $x = -1$ , there are exactly four solutions.*
- (iii) *If  $x > 0$  is a square, there are finitely many solutions, explicitly given by the divisor conditions (1.9).*
- (iv) *If  $x > 0$  is nonsquare, then there are either no solutions or infinitely many. In the latter case the solutions are a disjoint finite union of Pell orbits, one orbit for each seed in  $\mathcal{R}_x$ .*

*Proof.* Only the last assertion requires comment. The seed set is finite by (1.15); uniqueness in Proposition 4.2 makes the corresponding orbits disjoint; and Proposition 4.3 shows that every nonempty orbit is infinite.  $\square$

**Corollary 5.2** (An explicit infinite family). *Define positive integers  $Y_n, Z_n$  by*

$$(5.1) \quad (Y_0, Z_0) = (23, 156)$$

*and*

$$(5.2) \quad \begin{aligned} Z_{n+1} &= 24Z_n + 115Y_n, \\ Y_{n+1} &= 5Z_n + 24Y_n \end{aligned} \quad (n \geq 0).$$

*Then, for every  $n \geq 0$  and arbitrary signs  $\sigma, \tau \in \{1, -1\}$ ,*

$$(5.3) \quad (x, y, z) = (23, \sigma Y_n, \tau Z_n)$$

*is a solution of (1.1). These give infinitely many distinct integer solutions.*



*Proof.* The two identities

$$(5.4) \quad 24^2 - 23 \cdot 5^2 = 1$$

and

$$(5.5) \quad 156^2 - 23 \cdot 23^2 = 12169 = 23^3 + 2$$

show that

$$(5.6) \quad Z_n + Y_n\sqrt{23} = (156 + 23\sqrt{23})(24 + 5\sqrt{23})^n$$

has norm  $23^3 + 2$  for every  $n \geq 0$ . Equating coefficients after one further multiplication by  $24 + 5\sqrt{23}$  gives exactly the recurrence (5.2). Hence

$$Z_n^2 - 23Y_n^2 = 23^3 + 2,$$

which is (1.1) with  $x = 23$ . Moreover,

$$Y_{n+1} = 5Z_n + 24Y_n > Y_n,$$

so the pairs  $(Y_n, Z_n)$  are distinct. The first two are

$$(Y_0, Z_0) = (23, 156), \quad (Y_1, Z_1) = (1332, 6389).$$

□

*Remark 5.3* (The local condition is not sufficient). Membership in  $\mathcal{A}$  is a necessary restriction on a positive value of  $x$ , but it is not sufficient. For example,  $7 \in \mathcal{A}$ , yet  $x = 7$  gives no solution. Indeed, a solution would satisfy

$$(5.7) \quad z^2 - 7y^2 = 7^3 + 2 = 345.$$

Modulo 5, this becomes  $z^2 \equiv 2y^2 \pmod{5}$ . If  $5 \nmid y$ , then  $(zy^{-1})^2 \equiv 2 \pmod{5}$ , impossible because the square residues modulo 5 are 0, 1, 4. Thus  $5 \mid y$ , and then  $5 \mid z$  as well. Consequently 25 divides the left-hand side of (5.7), whereas  $25 \nmid 345$ , a contradiction. The finite seed condition  $\mathcal{R}_D \neq \emptyset$  is the exact remaining global condition in the nonsquare case.